

A crossed random-effects latent class model for annotation ensembles of large language models

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Abstract

Large language models (LLMs) are increasingly used as annotators, but every annotation is produced by a configuration of model, prompt and stochastic decoding run, and configurations that share training corpora and alignment procedures do not err independently. [ABSTRACT: 150–250 words, no citations; to be finalised with simulation results.]

Keywords: latent class model, crossed random effects, Dawid–Skene, large language models, text annotation, conditional dependence

MSC Classification: 62H30 , 62F15 , 62J12 , 62P25 , 68T50

1 Introduction

Large language models (LLMs) are now used routinely to label text at a scale and cost that human coding cannot match, in political science, management research and public administration among other fields (Gilardi et al. 2023; Carlson and Burbano 2026). Each label, however, is not the output of “the model” but of a *configuration*: a particular model, a particular prompt and a particular stochastic decoding run. Semantically equivalent prompts change accuracy by tens of percentage points (Barrie et al. 2024; Chen et al. 2026), nominally deterministic decoding is not deterministic (Atil et al. 2024), and the choice of model, prompt and temperature can flip the substantive conclusions of a downstream analysis (Baumann et al. 2025). Treating LLM output as a fixed measurement therefore hides a large and structured source of error.

The natural response is to annotate every item with several configurations and aggregate. The dominant aggregation model, due to Dawid and Skene (1979), treats each annotator as an independent noisy channel with its own class-conditional error

rates and estimates true labels and error rates jointly without a gold standard. Its assumption of conditional independence given the true class was reasonable for human coders working separately. It is not reasonable for an LLM ensemble: configurations that share a model share its representation of the text; configurations that share a prompt share its reading of the task; and all configurations share the training corpora and alignment procedures of the model families that produced them. The diagnostic-testing literature learned decades ago that ignoring such dependence overstates the accuracy of the individual tests and biases prevalence estimates (Vacek 1985; Qu et al. 1996).

This paper proposes a latent class model in which the dependence between LLM annotations is not assumed away, nor left arbitrary, but derived from the factorial design of the annotation ensemble. Conditional on the true class, the log-odds of a positive label is decomposed into class-specific model and prompt effects plus three item-level random effects: an item-ambiguity effect shared by every configuration, an item $\times$ model effect and an item $\times$ prompt effect; repeated decoding runs are conditionally independent given these effects. We call this the crossed random-effects latent class model (CRE-LCM). It nests the Dawid–Skene model and the one-factor random-effects model of Qu et al. (1996) as special cases, and it returns, in addition to posterior class probabilities, a decomposition of annotation noise into item, model, prompt and run components on a common scale.

Our contributions are threefold. First, we show by simulation that under crossed dependence of realistic size the Dawid–Skene estimator overstates specificity and inflates prevalence, and that the bias grows monotonically with the strength of dependence. Second, we analyse the identifiability of the CRE-LCM. The crossed item $\times$ model and item $\times$ prompt effects are identified by the replication structure of the design, but the shared item-ambiguity effect competes with the latent class itself and produces a second, merged-class posterior mode. We compare five remedies and show that the mode is avoided by initialising the sampler at the Dawid–Skene solution, and removed altogether by anchoring a small random subset of items with known labels. Third, we quantify, over a grid of dependence levels, sample sizes and numbers of prompts, how CRE-LCM compares with majority vote and Dawid–Skene in prevalence bias, accuracy and calibration of posterior labels, interval coverage and recovery of the variance shares, including a misspecified data-generating process with heavy-tailed random effects.

Section 2 reviews the three literatures the paper connects. Section 3 defines the model, its interpretation and its estimation, and presents the identifiability analysis. Section 4 reports the simulation study. Section 5 outlines the application and Section 6 discusses design implications and limitations.

2 Background and related work

2.1 Aggregating noisy labels without a gold standard

Dawid and Skene (1979) modelled each rater by a class-conditional confusion matrix and estimated true labels and rater accuracies jointly by the EM algorithm. The

model remains the foundation of crowdsourcing and unsupervised ensemble aggregation. Extensions add item difficulty (?), Bayesian priors and hierarchical structure, and model-based agreement coefficients (Warrens 2010, 2013); latent class models with local dependence have been studied in this journal in the context of model selection and classification (Oberski et al. 2013; Marbac et al. 2016; Oberski 2016). In the LLM setting the usual recommendation is to treat each model, or each model–prompt pair, as an annotator and apply Dawid–Skene or a descendant (Camuffo et al. 2026). This mapping inherits the assumption of conditional independence given the true class, which LLM configurations violate in a structured way.

2.2 Conditional dependence in latent class models

The diagnostic-testing literature encountered the same violation: multiple tests applied to the same subject share unmeasured subject characteristics, so the conditionally independent latent class model fits poorly and biases accuracy estimates (Vacek 1985). Qu et al. (1996) introduced a subject-level Gaussian random effect with a class-specific loading, and gave diagnostics for residual dependence. Subsequent work compared log-linear and Gaussian random-effects formulations (Xu et al. 2019), studied crossed subject×rater random effects for ordinal ratings without a gold standard together with robustness to a misspecified random-effects distribution (Xie et al. 2013), and produced software (Beath 2017). In all of this work the rater is a single unit. In an LLM ensemble the rater has a known factorial structure, model×prompt×run, which allows the dependence to be specified from the design rather than assumed or estimated pairwise.

2.3 LLM annotation: variability, measurement error and remedies

Three strands are relevant. The first documents instability: prompt paraphrases that preserve meaning change classification accuracy substantially (Barrie et al. 2024; Chen et al. 2026), decoding is not deterministic even at nominally deterministic settings (Atil et al. 2024), and LLM judges agree with each other more than with task-specific human annotation (Bavaresco et al. 2025; Norman et al. 2026). The second traces the consequences downstream: Baumann et al. (2025) replicated 37 annotation tasks with 18 models and showed that implementation choices produce Type I, II, S and M errors; Egami et al. (2024) and Camuffo et al. (2026) formalise the measurement-error argument, showing that annotation error correlated with covariates yields inconsistent regression estimates regardless of average accuracy, and that model choice can account for the bulk of annotation variance; Messing (2026) extend this to evaluation pipelines and Carlson and Burbano (2026) give guidelines for management research. The third strand proposes statistical remedies. Zhang and Martinez (2025) reframe LLM stochasticity as measurement error and propose a Bayesian generalised-linear version of Dawid–Skene with covariates on the latent state; their raters remain conditionally independent. Chen et al. (2026) take the opposite route: because LLMs share corpora and alignment their errors may be arbitrarily dependent, so point identification is abandoned in favour of bounds on prevalence. Prediction-powered

inference (Angelopoulos et al. 2023) offers a design-based route that requires a labelled subsample. Variance-aware protocols (Camuffo et al. 2026) and error-decomposition frameworks (Xu et al. 2026; Nguyen et al. 2023) recommend prompt ensembles, replication and noise-aware aggregation but do not supply a model in which the dependence between configurations is estimated.

2.4 Positioning

The literature therefore offers independence-based aggregators known to be misspecified for LLM ensembles, bounds that discard the information in the ensemble’s design, and protocols without an accompanying inferential model. CRE-LCM occupies the middle ground: dependence is given the structure implied by how the annotations were generated. This yields point identification under conditions we state, interpretable variance components that translate into annotation-design decisions, and a natural home for the covariate and multiclass extensions of the existing approaches.

3 Model

3.1 Setting and notation

Items $i = 1, \dots, N$ have an unobserved binary class $c_i \in \{0, 1\}$ with prevalence $\pi_1 = \Pr(c_i = 1)$. Each item is annotated by every configuration (m, p, r) of M models, P prompts and R decoding runs, giving binary labels Y_{impr} . Runs are exchangeable within a model–prompt pair, so the sufficient statistic is the count $S_{imp} = \sum_r Y_{impr} \in \{0, \dots, R\}$.

3.2 The crossed random-effects latent class model

Conditional on the class and on three item-level random effects, the labels are independent Bernoulli variables with

$$\text{logit } \Pr(Y_{impr} = 1 \mid c_i = k, \theta_i, \phi_{im}, \psi_{ip}) = \mu_k + a_{mk} + b_{pk} + \theta_i + \phi_{im} + \psi_{ip}, \quad (1)$$

where $\mu_0 < \mu_1$ are class intercepts (the ordering identifies the labels), a_{mk} are fixed model effects with $a_{1k} = 0$, $b_{pk} \sim N(0, \tau_k^2)$ are prompt effects, and

$$\theta_i \sim N(0, \sigma_\theta^2), \quad \phi_{im} \sim N(0, \sigma_\phi^2), \quad \psi_{ip} \sim N(0, \sigma_\psi^2) \quad (2)$$

are mutually independent. The effect θ_i is the ambiguity of item i as seen by every configuration; ϕ_{im} is the idiosyncratic reading of item i by model m , shared across the prompts and runs of that model; ψ_{ip} is the idiosyncratic reading induced by prompt p , shared across models and runs. Given the effects, $S_{imp} \mid c_i = k \sim \text{Binomial}(R, \text{expit}(\eta_{impk}))$ with η_{impk} the right-hand side of (1), and the marginal

likelihood of item i is

$$L_i = \sum_{k=0}^1 \pi_k \int \prod_{m=1}^M \prod_{p=1}^P \binom{R}{S_{imp}} \text{expit}(\eta_{impk})^{S_{imp}} (1 - \text{expit}(\eta_{impk}))^{R-S_{imp}} dF(\theta_i, \phi_i, \psi_i). \quad (3)$$

With $\sigma_\theta = \sigma_\phi = \sigma_\psi = 0$, (1) is the Dawid–Skene model with $M \times P$ raters whose sensitivities and specificities are $\text{expit}(\mu_1 + a_{m1} + b_{p1})$ and $1 - \text{expit}(\mu_0 + a_{m0} + b_{p0})$; with $\sigma_\phi = \sigma_\psi = 0$ it is the one-factor random-effects model of [Qu et al. \(1996\)](#) with a common loading. The binomial reduction over runs is exact because runs are conditionally independent given the effects; it is the only place where the run dimension enters.

3.3 Variance components

On the latent logistic scale the total variance of the label propensity of an item, over configurations and runs, decomposes as $\sigma_\theta^2 + \sigma_\phi^2 + \sigma_\psi^2 + \pi^2/3$, the last term being the variance of the standard logistic distribution that generates the run-level Bernoulli outcome. The shares

$$s_{\text{item}} = \frac{\sigma_\theta^2}{V}, \quad s_{\text{model}} = \frac{\sigma_\phi^2}{V}, \quad s_{\text{prompt}} = \frac{\sigma_\psi^2}{V}, \quad s_{\text{run}} = \frac{\pi^2/3}{V}, \quad V = \sigma_\theta^2 + \sigma_\phi^2 + \sigma_\psi^2 + \pi^2/3, \quad (4)$$

measure the fraction of annotation noise attributable to item ambiguity, to model idiosyncrasy, to prompt idiosyncrasy and to decoding stochasticity. They have a direct design reading: noise in the model and prompt components is reduced by adding models and prompts, noise in the run component by adding runs, and item ambiguity by none of these. A large s_{model} with a small s_{run} says that repeating the same configuration is wasted effort.

3.4 Identifiability

The model is identified up to the usual label permutation, which the ordering $\mu_0 < \mu_1$ removes, provided $M \geq 2$, $P \geq 2$ and $R \geq 2$: the item $\times$ model effect is identified because the same item is seen under P prompts within a model, the item $\times$ prompt effect because it is seen under M models within a prompt, and the run-level variance because the R runs replicate a fixed configuration. The shared effect θ_i is the delicate component. An item on which every configuration returns 1 is consistent both with $c_i = 1$ and with $c_i = 0$ and a large θ_i ; the two explanations differ only through the prior on θ_i and the class-specific intercepts. In finite samples this creates a second posterior mode in which the two class intercepts nearly coincide, the mixing proportion drifts towards the majority class and all heterogeneity is carried by θ_i . We call this the *merged-class* mode. Section ?? documents it and compares five remedies: (a) a class-specific scale $\sigma_{\theta,k}$ in the manner of [Qu et al. \(1996\)](#); (b) omitting θ_i ; (c) anchoring a small random subset of items with known labels; (d) an informative prior on σ_θ ; and (e) initialising the sampler at the Dawid–Skene solution.

3.5 Estimation

We fit the model in Stan (Carpenter et al. 2017) with the No-U-Turn sampler, marginalising the discrete class as in (3) and integrating the random effects by sampling them, in a non-centred parameterisation. Priors are weakly informative: $\pi_1 \sim \text{Beta}(1, 1)$, $\mu_k \sim \text{N}(0, 2^2)$, $a_{mk} \sim \text{N}(0, 1)$, half-normal(0, 0.5²) on τ_k and half-normal(0, 1) on σ_θ , σ_ϕ , σ_ψ . Posterior class probabilities $\Pr(c_i = 1 \mid \mathbf{S})$ are computed in the generated quantities from the same marginalisation. Following the analysis of Section 3.4, all chains are initialised at the Dawid–Skene solution (prevalence and the logits of mean sensitivity and one minus mean specificity; random effects at zero) and convergence is assessed with split- $\hat{R}$ and bulk and tail effective sample sizes (Vehtari et al. 2021), together with per-chain posterior means of π_1 and μ , which expose the merged-class mode directly. Leave-one-out cross-validation (Vehtari et al. 2017) is not usable for comparing variants here: each item carries $1 + M + P$ of its own random effects, so leaving an item out changes the posterior substantially and almost all Pareto k diagnostics exceed 0.7. The Stan programs are listed in Appendix A.

4 Simulation study

5 Application

6 Discussion

Supplementary information. Stan code for all model variants, additional tables and the full per-cell simulation results are provided as Online Resource 1. The code repository is available through an anonymised link for review.

Appendix A Stan code

```
// crossed_lcre.stan
// Binary latent class model with crossed random effects for
// LLM annotation ensembles (model x prompt x run).
// The discrete class c_i is marginalised out.

data {
  int<lower=1> N;           // items
  int<lower=1> M;           // LLM models
  int<lower=1> P;           // prompts
  int<lower=1> R;           // runs per (model, prompt)
  array[N, M, P] int<lower=0, upper=R> S; // # runs labelled 1
}

parameters {
  real<lower=0, upper=1> pi1; // prevalence of class 1
  ordered[2] mu;             // class intercepts (label
    identification)
  matrix[M - 1, 2] a_raw;    // model effects, model 1
    = reference
  matrix[P, 2] b_z;          // prompt effects (non-
    centred)
```

```

vector<lower=0>[2] tau_b;
real<lower=0> s_th; // item ambiguity SD
real<lower=0> s_ph; // item x model SD
real<lower=0> s_ps; // item x prompt SD
vector[N] th_z;
matrix[N, M] ph_z;
matrix[N, P] ps_z;
}

transformed parameters {
  matrix[M, 2] a;
  matrix[P, 2] b;
  a[1] = rep_row_vector(0, 2);
  if (M > 1) a[2:M] = a_raw;
  for (k in 1:2) b[, k] = tau_b[k] * b_z[, k];
}

model {
  // priors
  pi1 ~ beta(1, 1);
  mu ~ normal(0, 2);
  to_vector(a_raw) ~ normal(0, 1);
  to_vector(b_z) ~ std_normal();
  tau_b ~ normal(0, 0.5);
  s_th ~ normal(0, 1);
  s_ph ~ normal(0, 1);
  s_ps ~ normal(0, 1);
  th_z ~ std_normal();
  to_vector(ph_z) ~ std_normal();
  to_vector(ps_z) ~ std_normal();

  // likelihood, marginalising the latent class
  for (i in 1:N) {
    vector[2] lp = [log1m(pi1), log(pi1)]';
    for (m in 1:M) {
      for (p in 1:P) {
        real uu = s_th * th_z[i] + s_ph * ph_z[i, m] + s_ps * ps_z[i, p];
        for (k in 1:2)
          lp[k] += binomial_logit_lpmf(S[i, m, p] | R, mu[k] + a[m, k] + b[p, k] + uu);
      }
    }
    target += log_sum_exp(lp);
  }

  generated_quantities {
    vector[N] post1; // P(c_i = 1 | data)
    vector[N] log_lik; // pointwise (per item) log-likelihood for L00
    // variance shares on the latent logistic scale (run-level)
    residual = pi^2/3;
    real v_res = square(pi()) / 3;
    real v_tot = square(s_th) + square(s_ph) + square(s_ps) + v_res;
    real share_item = square(s_th) / v_tot;
    real share_model = square(s_ph) / v_tot;
    real share_prompt = square(s_ps) / v_tot;
    real share_run = v_res / v_tot;
  }
}

```

```

for(i in 1:N){
  vector[2] lp = [log1m(pi1), log(pi1)]';
  for (m in 1:M) for (p in 1:P) {
    real u = s_th * th_z[i] + s_ph * ph_z[i, m] + s_ps * ps_z[i, p];
    for (k in 1:2)
      lp[k] += binomial_logit_lpmf(S[i, m, p] | R, mu[k] + a[m, k] + b[p, k] + u);
  }
  log_lik[i] = log_sum_exp(lp);
  post1[i] = exp(lp[2] - log_lik[i]);
}

```

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